

# Architecture & security review package

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Diagrams and inventories for security/ATO review of the Claude apps gateway deployment (AWS GovCloud `us-gov-west-1`). Everything here is generated from the templates in `cloudformation/` at the commit this file ships with — when the code and this document disagree, the code wins and this file needs a PR.

The diagrams are hand-laid-out SVGs under [docs/ato/diagrams/](#), generated by [docs/ato/diagrams/generate.py](#) (edit the script, re-run it, and commit both). SVGs print cleanly and embed directly in submission documents; for PNG exports: `python3 -c "import cairosvg; cairosvg.svg2png(url='docs/ato/diagrams/<name>.svg', write_to='<name>.png', scale=2)"`.

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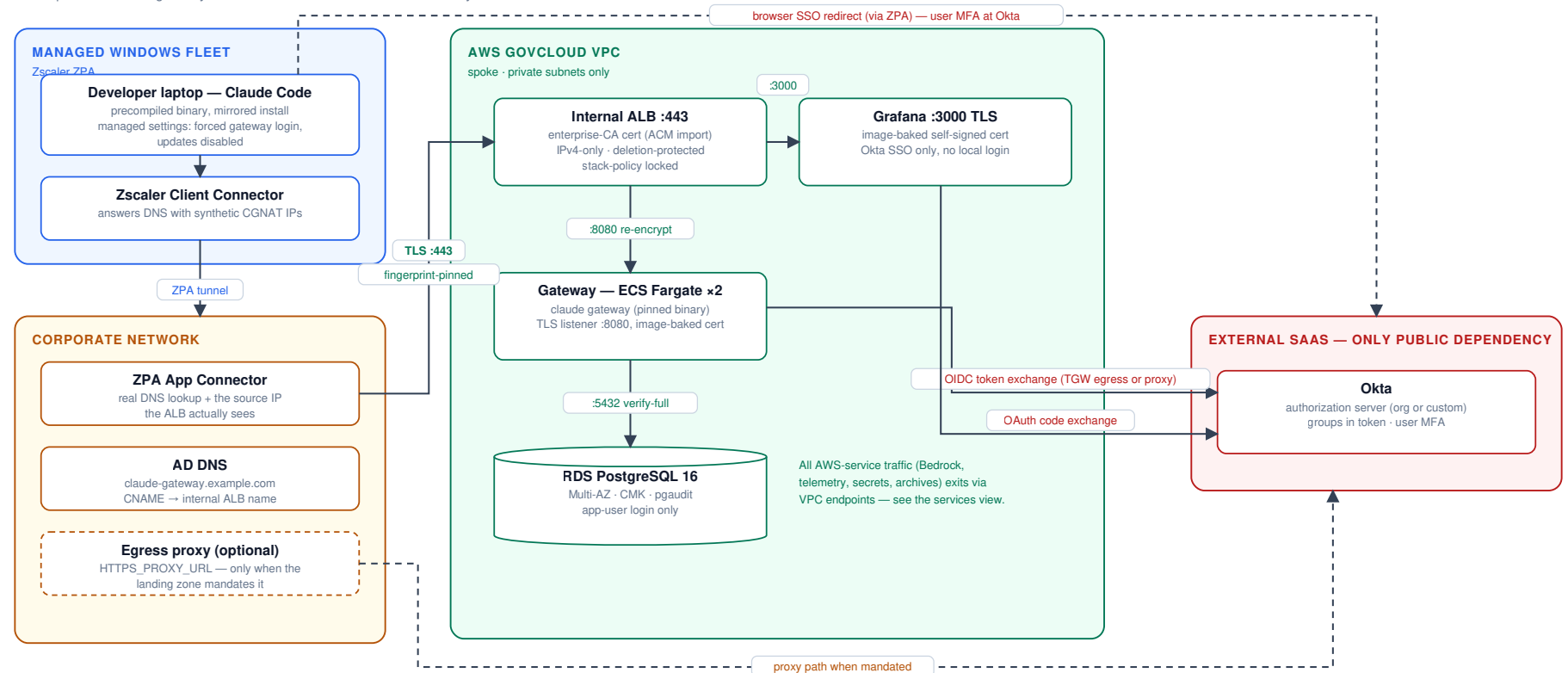
## 1. System architecture & trust boundaries

Four trust zones: the managed Windows fleet (Zscaler ZPA), the corporate network (App Connectors + AD DNS + egress proxy), the AWS GovCloud VPC (all workload components), and external SaaS (Okta — the single public dependency; Bedrock and all other AWS services are reached over private endpoints or the AWS backbone). Two complementary views; the combined single-page version is in the [appendix](#).

**Access & network path** — how requests reach the gateway and how the OIDC flows leave:

## Access & network path

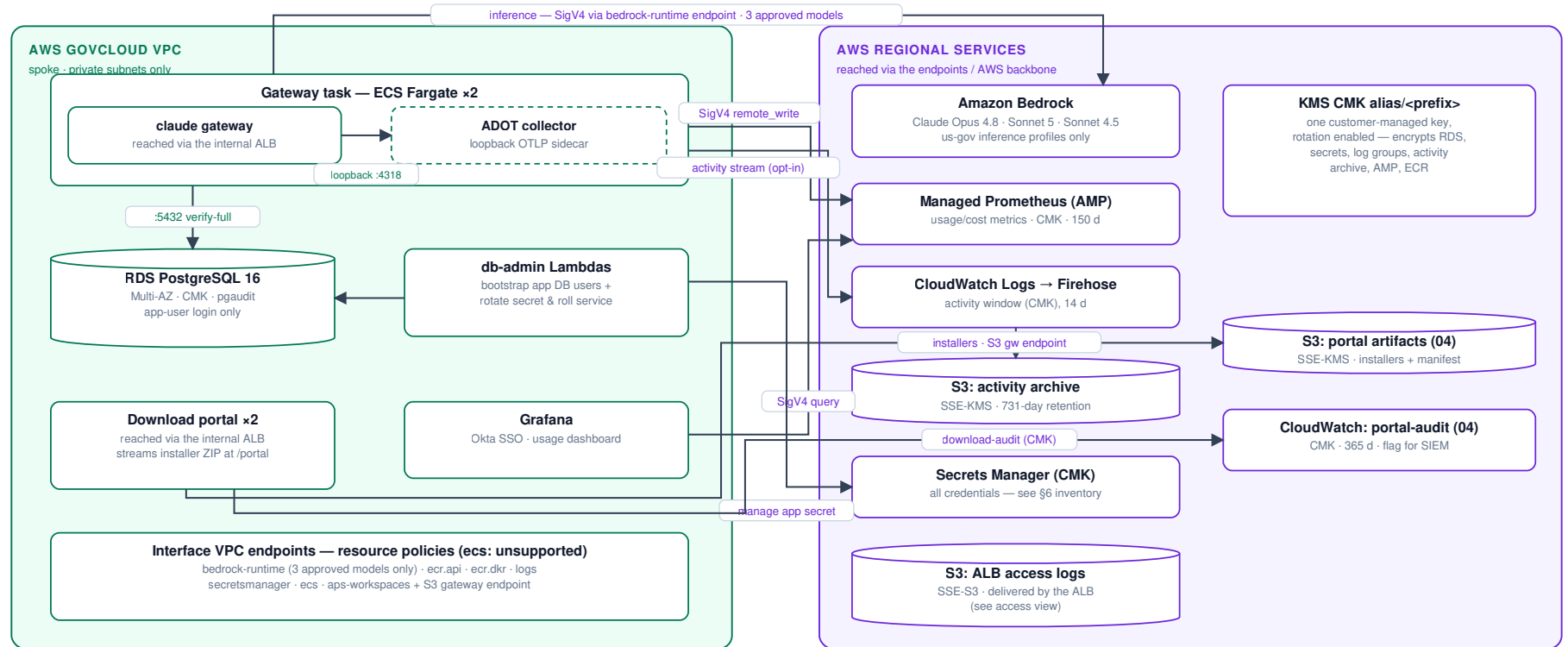
how requests reach the gateway and how the two OIDC flows leave — every arrow is TLS



**Workloads & AWS data services** — inference, telemetry, secrets and encryption inside AWS:

## Workloads & AWS data services

inference, telemetry, secrets and encryption — developer access path is in the access view



The ADOT collector is a co-resident loopback sidecar in the gateway task (OTLP over 127.0.0.1), so telemetry never crosses the network in the clear — there is no gateway-to-collector network hop. Okta egress and the developer path are on the access view.

Secrets are injected into tasks at launch by ECS (execution roles hold GetSecretValue + kms:Decrypt on exactly their own secrets — see §6).

Download portal (04): serves the installer ZIP from the CMK-encrypted artifacts bucket via the S3 gateway endpoint and writes a download-audit stream; its Okta OIDC egress is on the access view.

Key boundary statements for the reviewer:

- **No public ingress.** The ALB is internal; reaching it requires the ZPA app segment (or VPN) and an allow-listed connector source CIDR.
- **No public egress from the inference path.** Bedrock is reached via an interface endpoint whose policy allows exactly the three approved models; WebSearch is disabled on gateway sessions.
- **One public dependency:** the Okta issuer (OIDC has no VPC endpoint). Egress rides TGW central NAT or the mandated corporate proxy, port-scoped by security groups.

- **Client fleet never contacts Anthropic:** binaries are mirrored, checksum- and GPG-verified, and distributed from an internal share; update paths are disabled via the installer's user settings (and optionally the gateway's central push), and the mirror-only network path is the backstop.
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## 2. Network flows, ports & TLS state

Every hop, its port, protocol, and where the TLS session terminates.

## Network flows, ports & TLS state — every hop in the system

one row per flow; row 6 is an in-task loopback hop (never on the network) · rows 11–15 are the download portal (stack 04)



TLS termination points: developer→ALB terminates on the enterprise cert (developers pin its fingerprint); ALB→task hops terminate on self-signed certs baked into each image at build (ALBs do not validate target certs; rotated by image rebuild).

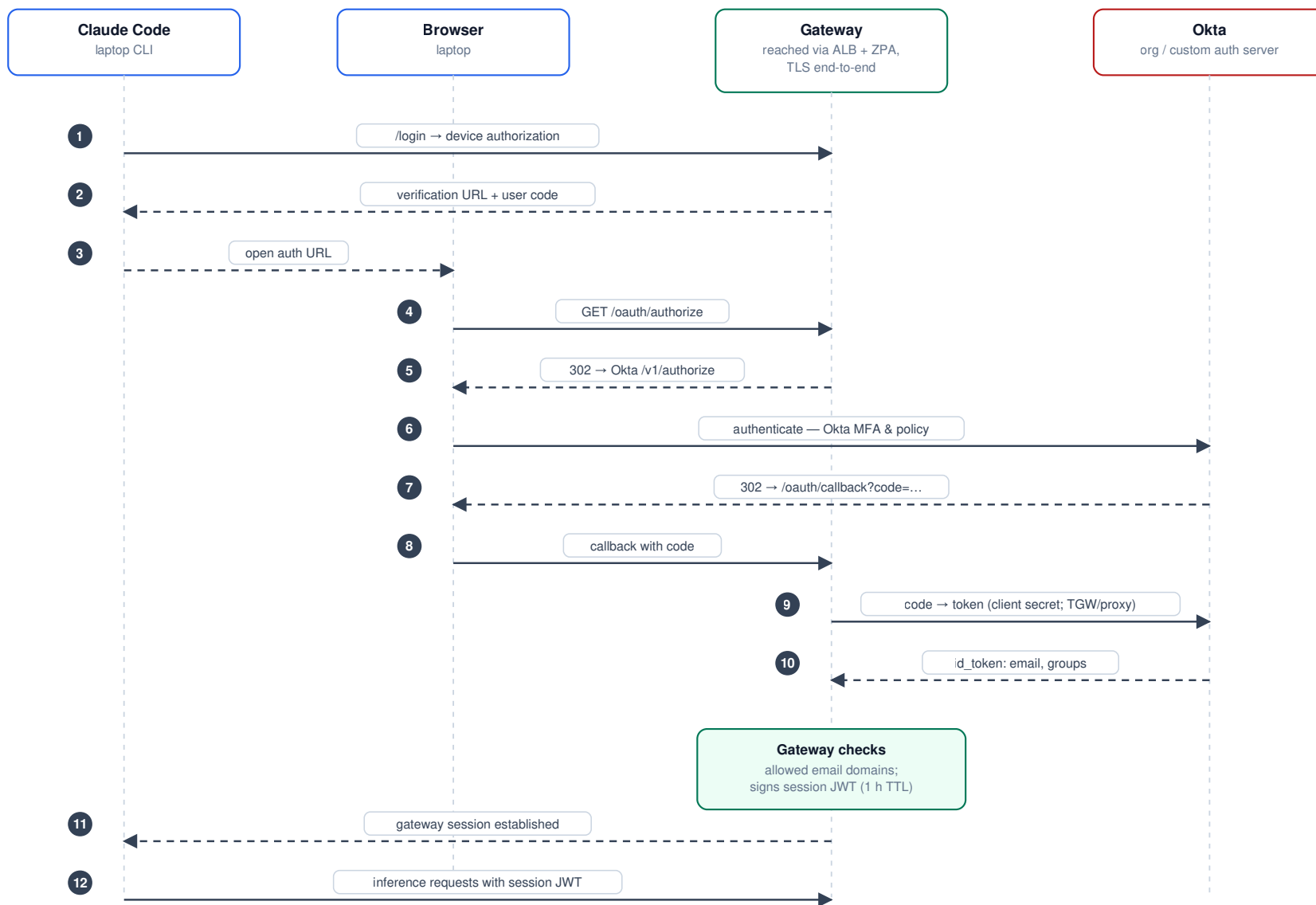
All AWS-service hops (5, 7, 8, 14, 15) also carry SigV4 request signing on top of TLS. DNS to the VPC resolver is exempt from security-group evaluation (AWS platform behavior).

| #    | Hop                             | Port      | Encryption                   | Server identity proven by                                                                                                                                                                                   |
|------|---------------------------------|-----------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Laptop → ALB                    | 443       | TLS (FIPS policy)            | Enterprise-CA cert; client pins SHA-256 fingerprint on first connect                                                                                                                                        |
| 2    | ALB → gateway                   | 8080      | TLS                          | Self-signed leaf generated on the build host and baked into the image (ALB does not validate targets; the key is identical across every task running that image and rotates only on the next image rebuild) |
| 3    | ALB → Grafana                   | 3000      | TLS                          | Self-signed leaf baked into the Grafana image (same build-time model)                                                                                                                                       |
| 4    | Gateway/Lambda → RDS            | 5432      | TLS <code>verify-full</code> | RDS regional CA bundle baked into images; <code>rds.force_ssl=1</code> server-side                                                                                                                          |
| 5    | Gateway → Bedrock               | 443       | TLS + SigV4                  | AWS cert; endpoint policy = 3 approved models only                                                                                                                                                          |
| 6    | Gateway → collector (sidecar)   | 4317–4318 | Loopback — no network hop    | Same Fargate task; the collector is a localhost sidecar with its receiver bound to <code>127.0.0.1</code> , so nothing off-task can reach it (see §10)                                                      |
| 7–8  | Gateway sidecar / Grafana → AMP | 443       | TLS + SigV4                  | AWS cert; endpoint policy = this workspace only                                                                                                                                                             |
| 9–10 | Gateway/Grafana → Okta          | 443       | TLS                          | Public CA; only internet-bound flow                                                                                                                                                                         |

### 3. Developer authentication (Okta OIDC)

## Developer authentication — Okta OIDC via the gateway

managed settings force gateway login; no local API keys exist



Audit identity chain: Okta identity → session JWT → user.id / user.email / user.groups stamped on every request and telemetry export. Grafana repeats this flow with its own client + strict group→role mapping.



Grafana uses the **same issuer** with its own client and redirect URI ( `/grafana/login/generic_oauth` ), strict Okta-group→role mapping ( `grafana-admins` → Admin; users in no mapped group are denied), PKCE, and the local login form disabled (break-glass `admin` requires a redeploy with `GRAFANA_DISABLE_LOGIN_FORM=false` ).

**Identity chain for audit:** Okta identity → gateway session JWT → every inference request and every telemetry export carries `user.id/user.email/user.groups` . Per-source-IP attribution is deliberately not relied on (ZPA collapses all users to connector IPs — documented in the README).

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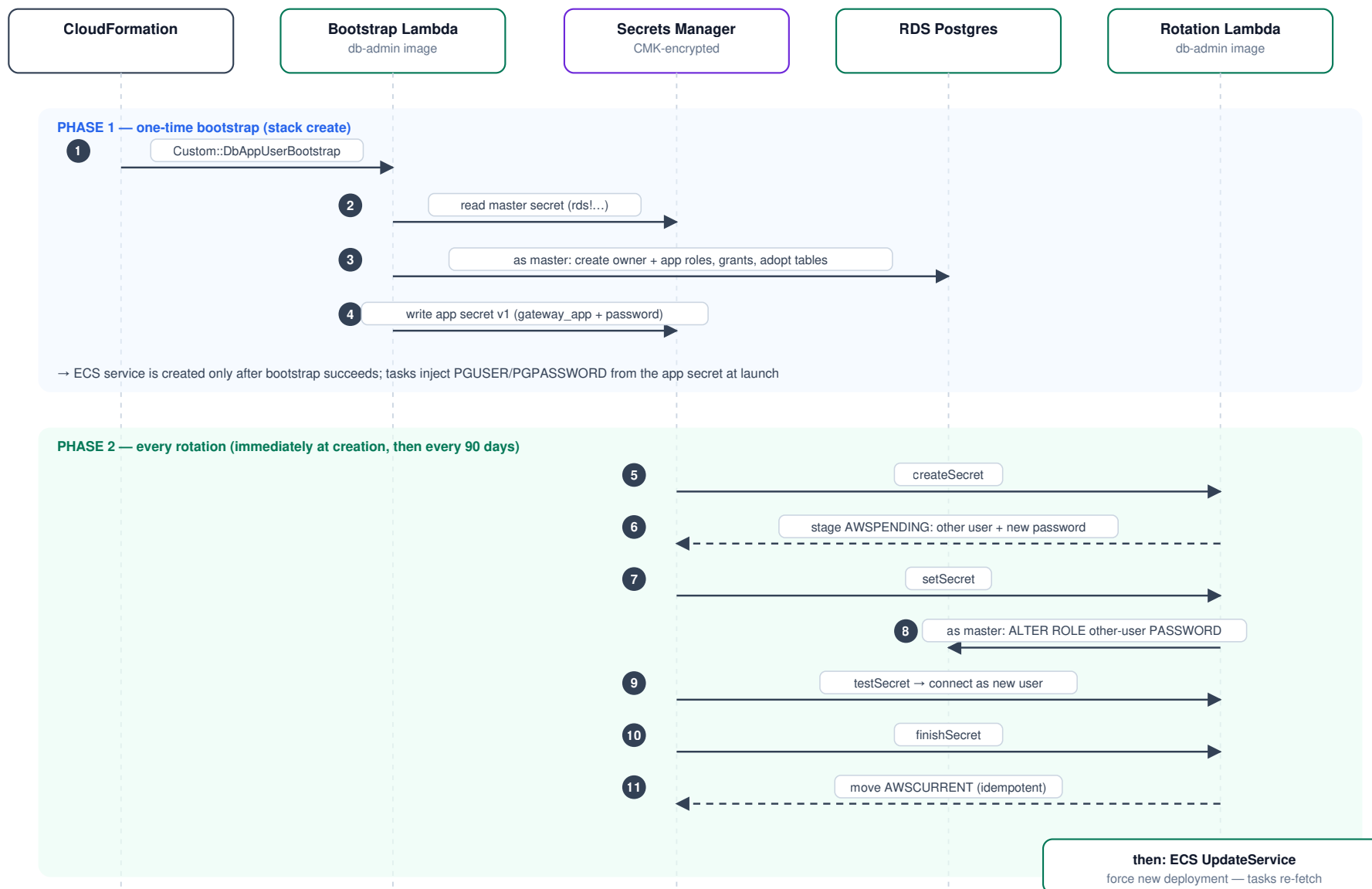
## 4. Database credential lifecycle

The gateway never holds the RDS master credential (AC-6). Identities:

| Postgres role                                             | Kind                     | Powers                                                                                                                | Used by                                      |
|-----------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <code>gw</code> (master)                                  | LOGIN, RDS-managed       | <code>rds_superuser</code> -adjacent                                                                                  | Break-glass humans + db-admin<br>Lambda only |
| <code>gateway_owner</code>                                | NOLOGIN                  | Owns schema objects; CONNECT/TEMPORARY/CREATE on the gateway DB only                                                  | Assumed at login by both app users           |
| <code>gateway_app</code> / <code>gateway_app_clone</code> | LOGIN (alternating pair) | Via <code>SET role</code> → <code>gateway_owner</code> ; <b>no CREATEROLE, no rds_superuser, cannot alter pgaudit</b> | Gateway tasks (exactly one is live)          |

## Database credential lifecycle — least privilege + alternating-users rotation

the gateway never holds the master credential; the master secret is break-glass only



Why alternating users: the PREVIOUS credential stays valid until the NEXT rotation, so running tasks never hold a dead password — the roll is never time-critical.

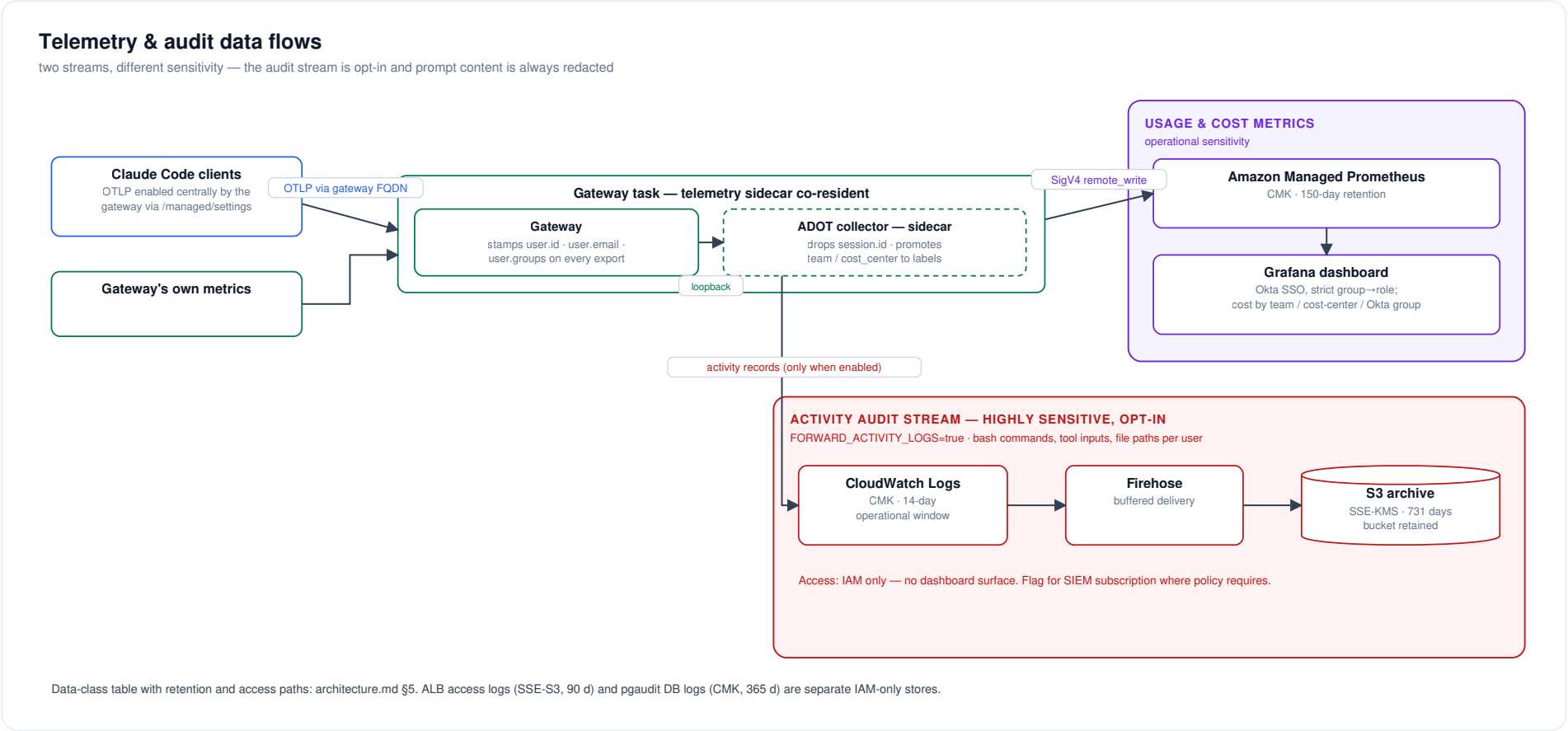
Failure handling: Secrets Manager retries any failed step; a persistently failing rotation trips the `<prefix>-db-rotation-errors` CloudWatch alarm.

Roles: gw (master, break-glass, RDS-rotated weekly) · gateway\_owner (NOLOGIN, owns schema) · gateway\_app / gateway\_app\_clone (LOGIN, SET role → owner; no CREATEROLE, no rds\_superuser, cannot touch pgaudit).

Failure containment: a failed rotation step is retried by Secrets Manager; a persistently failing rotation trips the `<prefix>-db-rotation-errors` CloudWatch alarm. The RDS master secret keeps its RDS-managed 7-day auto-rotation, which affects no running workload.

## 5. Telemetry & audit data flows

Two distinct streams with different sensitivity:



| Data class               | Contains                                                                  | Store                           | Retention                   | Access path                                                                              |
|--------------------------|---------------------------------------------------------------------------|---------------------------------|-----------------------------|------------------------------------------------------------------------------------------|
| Usage metrics            | tokens, cost, sessions, LoC, model, user identity, team/cost-center       | AMP (CMK)                       | 150 d (AMP fixed)           | Grafana via Okta SSO, role-mapped                                                        |
| Activity stream (opt-in) | bash commands, tool inputs, file paths, per user; prompt content redacted | CloudWatch (CMK) → S3 (SSE-KMS) | 14 d window / 731 d archive | IAM only; flagged for SIEM subscription                                                  |
| ALB access logs          | source connector IPs, URLs, timings                                       | S3 (SSE-S3)                     | 90 d                        | IAM only; SQL-searchable via the optional Athena stack (05) — results land CMK-encrypted |
| DB audit (pgaudit)       | DDL, role changes, writes (no bind values)                                | RDS → CloudWatch (CMK)          | 365 d group                 | IAM only                                                                                 |
| Session/spend store      | sessions, spend counters                                                  | RDS (CMK)                       | live                        | app DB user only                                                                         |

## 6. Secrets & key inventory

One customer-managed KMS key ( `alias/<prefix>`, rotation enabled, created by the DB stack or bring-your-own) encrypts everything below except where noted.

| Secret                                            | Store / name                                                                                                                | Created by                                                                      | Consumed by                                                                                                                                                                                                              | Rotation                                                                                                                                                    |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RDS master ( gw )                                 | Secrets Manager <code>rds!...</code>                                                                                        | RDS-managed                                                                     | <b>Humans (break-glass)</b> + db-admin Lambda                                                                                                                                                                            | RDS-managed, 7 d auto                                                                                                                                       |
| App DB user                                       | <code>&lt;prefix&gt;/db-app-user</code>                                                                                     | Bootstrap Lambda                                                                | Gateway tasks (ECS injection at launch)                                                                                                                                                                                  | Stack's rotation Lambda, 90 d default, alternating users + service roll                                                                                     |
| Okta client secret (gateway)                      | <code>&lt;prefix&gt;/oidc-client-secret</code>                                                                              | Stack (placeholder) → <code>set-okta-secret.sh</code>                           | Gateway tasks                                                                                                                                                                                                            | Manual, in Okta + script (rolls service)                                                                                                                    |
| Okta client secret (Grafana)                      | <code>&lt;prefix&gt;/grafana-oidc-client-secret</code>                                                                      | Stack (placeholder) → <code>set-grafana-oidc-secret.sh</code>                   | Grafana task                                                                                                                                                                                                             | Manual, same pattern                                                                                                                                        |
| Okta client secret (portal)                       | <code>&lt;prefix&gt;/portal-oidc-client-secret</code>                                                                       | Stack 04 (placeholder) → <code>set-portal-oidc-secret.sh</code>                 | Portal tasks                                                                                                                                                                                                             | Manual, same pattern                                                                                                                                        |
| JWT session-signing secret                        | <code>&lt;prefix&gt;/jwt-secret</code>                                                                                      | Stack (generated)                                                               | Gateway tasks                                                                                                                                                                                                            | Manual runbook: prepend → roll → remove                                                                                                                     |
| Portal session-signing secret                     | <code>&lt;prefix&gt;/portal-session-secret</code>                                                                           | Stack 04 (generated)                                                            | Portal tasks                                                                                                                                                                                                             | Manual (regenerate + roll)                                                                                                                                  |
| Grafana admin password                            | <code>&lt;prefix&gt;/grafana-admin-password</code>                                                                          | Stack (generated)                                                               | Break-glass only (login form disabled)                                                                                                                                                                                   | Manual                                                                                                                                                      |
| Spend-admin write key                             | <code>&lt;prefix&gt;/spend-admin-write-key</code>                                                                           | Stack (generated)                                                               | Gateway task only (config's <code>admin: block</code> recognizes it as the write-scope bearer token); break-glass via <code>set-spend-limit.sh</code> — no other task is ever handed this key                            | Manual, not automatic ( <code>GenerateSecretString</code> at create); procedure in <code>cost-controls.md</code> §7                                         |
| Spend-admin read key                              | <code>&lt;prefix&gt;/spend-admin-read-key</code>                                                                            | Stack (generated)                                                               | Gateway task ( <code>admin: block</code> ) <b>and</b> the download portal (04, imported via the <code>&lt;prefix&gt;-spend-read-key-arn</code> export) for <code>/portal/me</code> self-usage and the admin listing view | Manual, not automatic; rolling it means re-deploying both the gateway and portal services ( <code>cost-controls.md</code> §7)                               |
| TLS: enterprise leaf + key                        | ACM import                                                                                                                  | Enterprise CA via <code>import-enterprise-cert.sh</code>                        | ALB listener                                                                                                                                                                                                             | Manual re-import; <code>DaysToExpiry</code> alarm at 30 d                                                                                                   |
| TLS: image-baked certs (gateway, Grafana, portal) | Self-signed leaf generated with <code>openssl</code> on the build host, then COPY -ed into the image at <code>docker</code> | <code>build-and-push-image.sh</code> / <code>build-and-push-grafana.sh</code> / | ALB target connections (hops 2–3 in §2)                                                                                                                                                                                  | Every image rebuild, not per task launch; the key is an ECR image-layer artifact, identical across every task running that image tag until the next rebuild |

| Secret | Store / name                                     | Created by               | Consumed by | Rotation |
|--------|--------------------------------------------------|--------------------------|-------------|----------|
|        | build time — the image itself carries no openssl | build-and-push-portal.sh |             |          |

Handling rules embedded in tooling: secret values never appear on argv ( `file://` + mode-600 temp files), never in CloudFormation parameters, and the two `set-* -secret.sh` scripts roll their consuming service.

## 7. Security-group matrix

Default egress is removed from **every** group (inline rules replace the allow-all); the table is the complete connectivity graph.

**Deep dive with diagrams:** [network-access-controls.md](#) maps every SG rule pair visually (diagram 7), inventories each group (diagram 8), and traces one API call through all five enforcement layers - SGs, endpoint SG, endpoint policies, IAM, KMS (diagram 9).

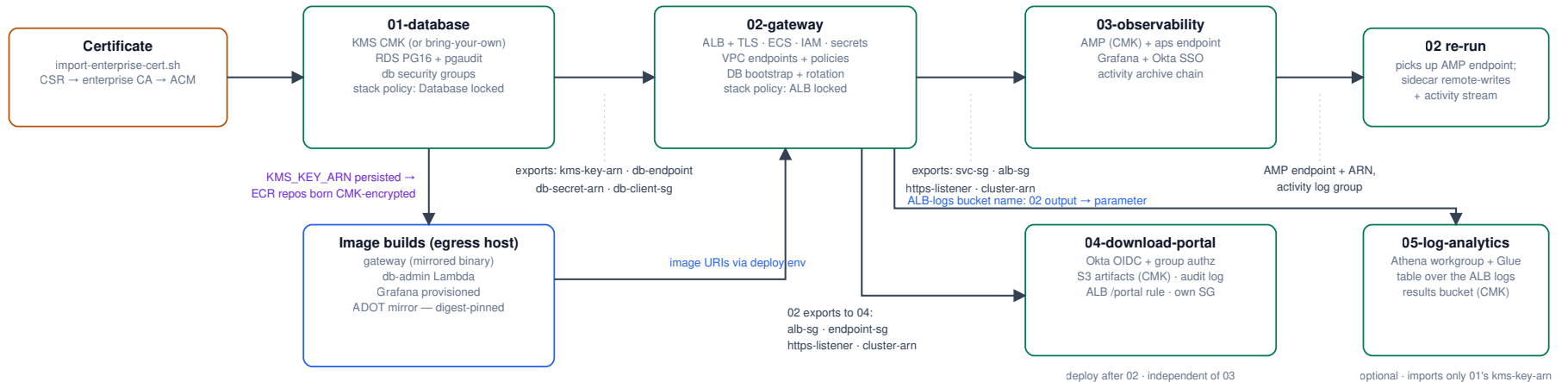
| SG                                                    | Ingress                                                                  | Egress                                                                                                                                                                                                                            |
|-------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alb                                                   | 443 from CLIENT_INGRESS_CIDR (ZPA connector subnets)                     | 8080 → svc ; 3000 → grafana (added by 03); 8080 → portal (added by 04); otherwise none                                                                                                                                            |
| svc (gateway tasks, incl. the ADOT collector sidecar) | 8080 from alb                                                            | 443 → 0.0.0.0/0 (Okta + AWS via endpoints/TGW; the sidecar's AMP/CloudWatch traffic rides this SG); proxy port when configured; 443 → amp-endpoint (added by 03, for the sidecar's AMP remote-write); 5432 via attached db-client |
| db-client (attached to tasks + db-admin Lambdas)      | —                                                                        | 5432 → db only                                                                                                                                                                                                                    |
| db                                                    | 5432 from db-client                                                      | none                                                                                                                                                                                                                              |
| db-admin (Lambdas)                                    | —                                                                        | 443 → 0.0.0.0/0 (Secrets Manager, ECS APIs)                                                                                                                                                                                       |
| grafana                                               | 3000 from alb                                                            | 443 → 0.0.0.0/0 (AMP, Okta); proxy port when configured                                                                                                                                                                           |
| portal (download portal, 04)                          | 8080 from alb                                                            | 443 → 0.0.0.0/0 (Okta, S3, CloudWatch); proxy port when configured                                                                                                                                                                |
| endpoints / amp-endpoint                              | 443 from svc / db-admin (resp. svc (gateway sidecar) + grafana + portal) | none                                                                                                                                                                                                                              |

DNS to the VPC resolver is exempt from SG evaluation (AWS platform behavior) and is why no port-53 rules appear.

## 8. Stack dependencies & deploy order

### Stack dependencies & deploy order

arrows carry CloudFormation exports — while imported, an export's value is locked (day-one decisions called out)



Locks a reviewer should know: the RDS storage CMK is fixed at creation (plus 01 ↔ 02 export locks) — a day-one decision; 03 and 04 must be deleted before 02 replacement-updates, 05 before 01; the ALB and Database carry stack policies denying Update:Replace / Update:Delete. Full teardown/update ordering: README — Teardown & update order.

Export locks (documented in README "Teardown & update order"): while a downstream stack imports an export, the upstream stack cannot change its value — most notably the RDS storage CMK is a **day-one decision**, and 03 must be deleted before 02 replacement-updates. The optional log-analytics stack (05: Athena + Glue over the ALB access logs) imports only 01's `ALBLogsBucketName` stack output and passes it as a parameter — deliberately not an export, so 02 gains no new export lock. The ALB and Database additionally carry CloudFormation **stack policies** denying `Update:Replace` / `Update:Delete`, so an accidental template change that would recreate them (new ALB DNS name → client DNS resubmission; new empty database) fails fast.

## 9. Data-at-rest encryption map

| Store                                                                                                                                                                             | Encryption                                | Key                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RDS storage + snapshots                                                                                                                                                           | SSE                                       | CMK                                                                                                                                                                                                            |
| RDS master secret / all Secrets Manager secrets                                                                                                                                   | SSE                                       | CMK                                                                                                                                                                                                            |
| CloudWatch log groups (ECS gateway [+ its collector sidecar, stream prefix <code>otel</code> ], grafana/portal, activity, portal-audit, RDS postgresql/pgaudit, db-admin Lambdas) | SSE                                       | CMK — every group is template-declared (never service-auto-created) precisely so the CMK and retention apply; all carry <code>DeletionPolicy: Retain</code> , so no log group is destroyed by a stack teardown |
| Activity archive bucket                                                                                                                                                           | SSE-KMS + bucket key                      | CMK                                                                                                                                                                                                            |
| Portal artifacts bucket (04)                                                                                                                                                      | SSE-KMS + bucket key                      | CMK                                                                                                                                                                                                            |
| Athena query-results bucket (05, optional)                                                                                                                                        | SSE-KMS + bucket key (workgroup-enforced) | CMK                                                                                                                                                                                                            |
| AMP workspace                                                                                                                                                                     | SSE                                       | CMK ( <code>ENCRYPT_AMP_WITH_CMK</code> , creation-time)                                                                                                                                                       |
| ECR repositories                                                                                                                                                                  | SSE-KMS at creation                       | CMK (when created after 01)                                                                                                                                                                                    |
| ALB access-logs bucket                                                                                                                                                            | SSE-S3 (AES-256)                          | <b>AWS-managed — ELB log delivery does not support KMS</b>                                                                                                                                                     |

## 10. Known accepted risks / review notes

Items a reviewer should see up front, with rationale (finding-by-finding detail in `security-assessment-2026-07.md`):

1. **There is no gateway→collector OTLP network hop** (§2 hop 6). The gateway refuses a non-HTTPS telemetry forward URL unless the host is localhost, and an internal-CA TLS listener would add a cert + CA the gateway must trust (an SC-17/SC-12 burden the org declined). The collector therefore runs as a **localhost sidecar** inside the gateway task: the gateway forwards over loopback ( `http://localhost:4318` ), so SC-8 is met by absence of transmission rather than by encryption. This requires `CLAUDE_GATEWAY_ALLOW_LOOPBACK=1` on the gateway container (its SSRF guard blocks loopback by default); the override is gated on telemetry being enabled and re-permits only loopback — EC2 IMDS and link-local addresses stay blocked. The gateway task role is the telemetry writer to AMP + the activity log group (scoped to this workspace and this log group).
2. **ALB access logs cannot use the CMK** — AWS platform limitation; SSE-S3 with public access blocked and lifecycle expiry.
3. **S3 Object Lock deferred** by decision; revisit if AU-9 WORM retention is mandated.
4. **Egress 443 to 0.0.0.0/0 from tasks** — required for Okta (public SaaS, no VPC endpoint, IP ranges not pinnable); mitigated by port scoping, endpoint policies for all AWS traffic, and the central inspection layer in the landing zone.
5. **Foundation-model IAM region wildcard** ( `bedrock::*:foundation-model/ <exact-model>` ) — required because GovCloud geo inference profiles fan out across us-gov regions; model IDs are exact.
6. **First app-secret rotation is asynchronous** — the stack goes green regardless; verify via the db-rotation log group / errors alarm after first deploy.



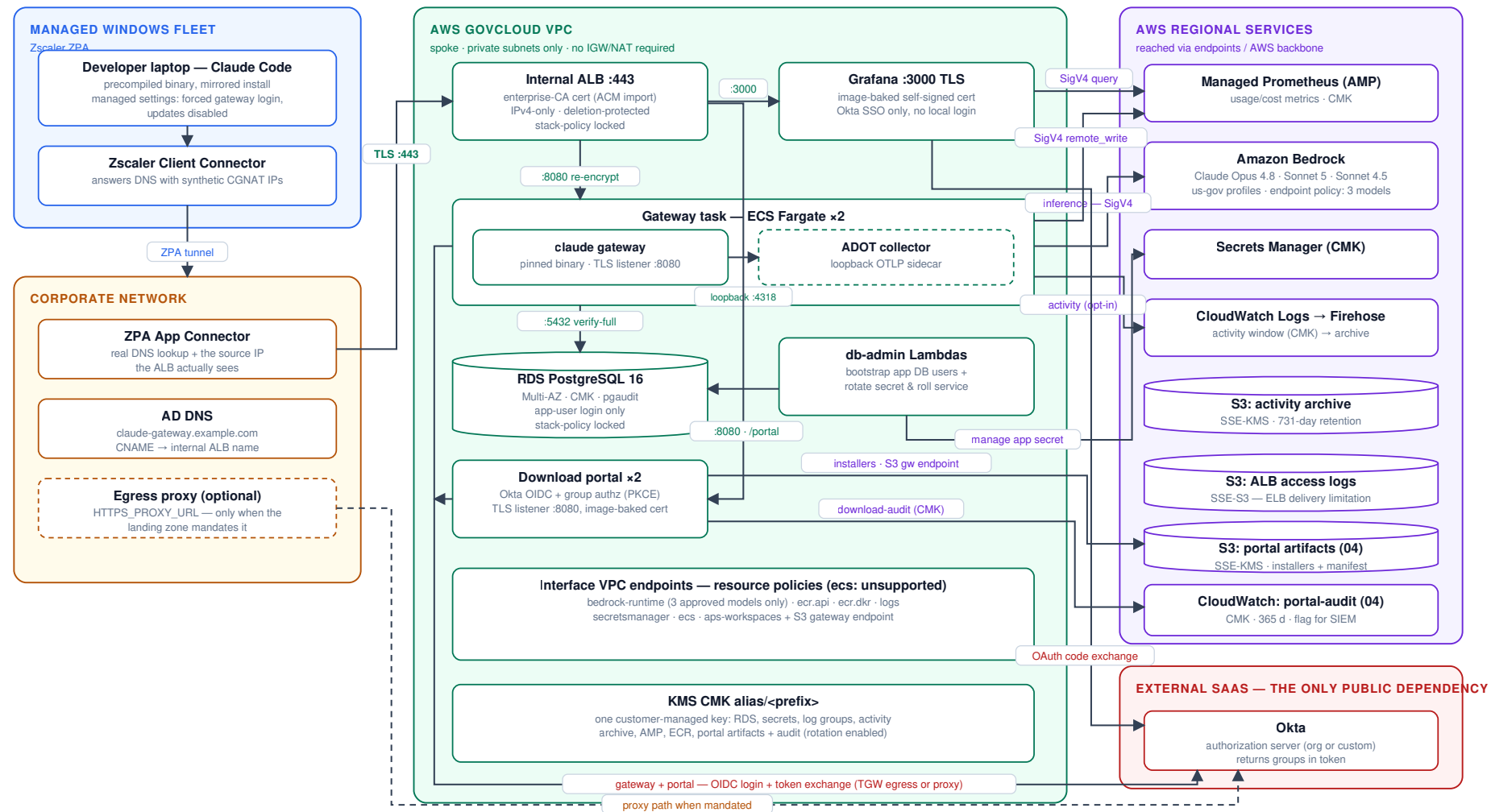
7. **The ALB→task TLS key (§2 hops 2–3, §6) is an image-build artifact, not a per-task secret.** It is generated once on the build host and baked into the gateway/Grafana/portal images; every task running a given image tag presents the identical key, and any principal able to pull that image (ECR read access) can extract it. This only protects the ALB-to-target hop — the ALB does not validate target certificates, and client trust is anchored to the enterprise leaf on the ALB listener, not this key — so the exposure is bounded to that hop; rotation is achieved by rebuilding and re-pushing the image under a new tag.
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## Appendix — combined system view

Everything from §1 on a single page — denser than the split views above, useful when a reviewer wants the whole system at a glance.

## System architecture & trust boundaries

Claude apps gateway in AWS GovCloud us-gov-west-1 — every arrow is TLS unless flagged



Boundary facts: no public ingress (internal ALB behind ZPA); no public egress from the inference path (Bedrock endpoint, 3-model policy); clients never contact Anthropic (mirrored, verified binaries; updates disabled).